Project Title: E-Commerce Customer Churn Analysis

Data Cleaning

Handling Missing Values and Outliers:

1)Impute mean for the following columns, and round off to the nearest integer ifmrequired: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

Ans) WarehouseToHome

step1: SELECT * FROM customer_churn;

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegister	PreferedOrderCat	SatisfactionScore	MaritalStatus
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3	Single
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3	Single
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single
50005	1	0	Phone	1	12	CC	Male	NULL	3	Mobile	5	Single
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5	Single
50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2	Divorced
50008	1	NULL	Phone	1	6	CC	Male	3	3	Mobile	2	Divorced
50009	1	13	Phone	3	9	E wallet	Male	NULL	4	Mobile	3	Divorced
50010	1	NULL	Phone	1	31	Debit Card	Male	2	5	Mobile	3	Single
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3	Divorced
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion	3	Single
50013	1	0	Phone	1	11	COD	Male	2	3	Mobile	3	Single
50014	1	0	Phone	1	15	CC	Male	3	4	Mobile	3	Divorced
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion	2	Single
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Step 2: SET SQL_SAFE_UPDATES = 0

UPDATE customer_churn SET WarehouseToHome =(SELECT Temp.AvgWTH FROM(SELECT ROUND(AVG(WarehouseToHome)) AS AvgWTH FROM customer_churn) AS Temp)WHERE WarehouseToHome IS NULL;

SELECT customerID, WarehouseToHome FROM customer_churn; ❼(updated WarehouseToHome)

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegister	PreferedOrderCat	SatisfactionScore	MaritalStatus
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3	Single
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3	Single
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single
50005	1	0	Phone	1	12	CC	Male	NULL	3	Mobile	5	Single
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5	Single
50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2	Divorced
50008	1	NULL	Phone	1	6	CC	Male	3	3	Mobile	2	Divorced
50009	1	13	Phone	3	9	E wallet	Male	NULL	4	Mobile	3	Divorced
50010	1	NULL	Phone	1	31	Debit Card	Male	2	5	Mobile	3	Single
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3	Divorced
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion	3	Single
50013	1	0	Phone	1	11	COD	Male	2	3	Mobile	3	Single
50014	1	0	Phone	1	15	CC	Male	3	4	Mobile	3	Divorced
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion	2	Single
50016	1	NULL	Phone	2	12	UPI	Male	3	3	Mobile	5	Married
50017	1	0	Computer	1	12	Debit Card	Female	NULL	4	Mobile	2	Single

HourSpendOnApp

select customerId,HourSpendOnApp from customer_churn where HourSpendOnApp is null;

customerId	HourSpendOnApp
50005	NULL
50009	NULL
50017	NULL
50038	NULL
50045	NULL
50046	NULL
50049	NULL
50058	NULL
50061	NULL
50079	NULL
50112	NULL
50118	NULL

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UPDATE customer_churn SET HourSpendOnApp =(SELECT Temp.Avg_HSOA FROM(SELECT ROUND(AVG(HourSpendOnApp)) AS Avg_HSOA FROM customer_churn) AS Temp)WHERE HourSpendOnApp IS NULL;

select * from customer_churn;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegister	PreferedOrderCat	SatisfactionScor
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Phone	1	12	CC	Male	3	3	Mobile	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1	NULL	Phone	1	6	CC	Male	3	3	Mobile	2
50009	1	13	Phone	3	9	E wallet	Male	3	4	Mobile	3
50010	1	NULL	Phone	1	31	Debit Card	Male	2	5	Mobile	3
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3

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select customerId,HourSpendOnApp from customer_churn where HourSpendOnApp is null;

Result Grid | Filter Rows:

customerId	HourSpendOnApp

OrderAmountHikeFromlastYear

select customerId,OrderAmountHikeFromlastYear from customer_churn where OrderAmountHikeFromlastYear is null;

Result Grid | Filter Rows: | Export:

customerId	OrderAmountHikeFromlastYear
50011	NULL
50041	NULL
50066	NULL
50107	NULL
50109	NULL
50120	NULL
50133	NULL
50202	NULL
50268	NULL
50287	NULL
50349	NULL
50373	NULL
50444	NULL
50471	NULL
50592	NULL
50622	NULL
50625	NULL
50642	NULL
50646	NULL
50690	NULL
50727	NULL

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UPDATE customer_churn SET OrderAmountHikeFromlastYear =(SELECT Temp.Avg_OAHF FROM(SELECT ROUND(AVG(OrderAmountHikeFromlastYear)) AS Avg_OAHF FROM customer_churn) AS Temp)WHERE OrderAmountHikeFromlastYear IS NULL;

select customerId,OrderAmountHikeFromlastYear from customer_churn where OrderAmountHikeFromlastYear is null;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customerId	OrderAmountHikeFromlastYear

select * from customer_churn;

CustomerID	Chun	Tenure	Pref	Warehou	Gend	HourSpendOnApp	NumberOfDeviceRegister	PreferedOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	Cou
50001	1	4	M...	6	Fe...	3	3	Laptop & Accessory	2	Single	9	1	11	1
50002	1	NULL	P...	8	Male	3	4	Mobile	3	Single	7	1	15	0
50003	1	NULL	P...	30	Male	2	4	Mobile	3	Single	6	1	14	0
50004	1	0	P...	15	Male	2	4	Laptop & Accessory	5	Single	8	0	23	0
50005	1	0	P...	12	Male	3	3	Mobile	5	Single	3	0	11	1
50006	1	0	C...	22	Fe...	3	5	Mobile Phone	5	Single	2	1	22	4
50007	1	NULL	P...	11	Male	2	3	Laptop & Accessory	2	Divorced	4	0	14	0
50008	1	NULL	P...	6	Male	3	3	Mobile	2	Divorced	3	1	16	2
50009	1	13	P...	9	Male	3	4	Mobile	3	Divorced	2	1	14	0
50010	1	NULL	P...	31	Male	2	5	Mobile	3	Single	2	0	12	1
50011	1	4	M...	18	Fe...	2	3	Others	3	Divorced	2	0	16	9
50012	1	11	M...	6	Male	3	4	Fashion	3	Single	10	1	13	0
50013	1	0	P...	11	Male	2	3	Mobile	3	Single	2	1	13	2
50014	1	0	P...	15	Male	3	4	Mobile	3	Divorced	1	1	17	0
50015	1	9	M...	15	Male	3	4	Fashion	2	Single	2	0	16	0
50016	1	NULL	P...	12	Male	3	3	Mobile	5	Married	5	1	22	1
50017	1	0	C...	12	Fe...	3	4	Mobile	2	Single	2	1	18	1

DaySinceLastOrder

select customerId,DaySinceLastOrder from customer_churn where DaySinceLastOrder is null;

Result Grid | Filter Rows:

customerId	DaySinceLastOrder
50089	NULL
50098	NULL
50127	NULL
50147	NULL
50152	NULL
50183	NULL
50189	NULL
50217	NULL
50228	NULL
50296	NULL
50332	NULL
50334	NULL

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select * from customer_churn;----------❼ to view null value

CustomerID	Chun	Tenure	Pref	Warehou	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatu	Number	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
50082	0	0	P...	NULL	Male	2	2	Mobile	4	Married	2	1	16	0	2	2	123
50083	0	NULL	C...	24	Fe...	3	3	Mobile	2	Single	2	0	13	1	1	7	122
50084	0	13	C...	15	Male	3	4	Laptop ...	2	Divorced	10	0	14	1	1	2	149
50085	0	19	M...	6	Male	3	3	Fashion	5	Married	8	0	16	0	1	3	194
50086	1	0	M...	10	Male	2	5	Fashion	2	Divorced	2	0	11	7	9	8	204
50087	0	3	C...	6	Male	2	4	Fashion	3	Single	2	0	22	2	5	8	185
50088	0	0	C...	26	Fe...	2	3	Laptop ...	3	Single	2	0	14	1	1	7	200
50089	0	24	M...	14	Male	2	4	Fashion	4	Divorced	6	0	16	1	4	NULL	225
50090	1	0	P...	NULL	Fe...	2	3	Mobile	5	Single	6	1	11	1	1	4	127
50091	0	14	M...	12	Fe...	3	3	Fashion	2	Single	4	1	18	0	NULL	0	236
50092	1	0	C...	32	Fe...	2	4	Laptop ...	5	Single	6	0	15	0	1	1	167
50093	1	NULL	P	6	Male	2	3	Mobile	3	Single	1	0	14	0	1	5	121

UPDATE customer_churnSET DaySinceLastOrder =(SELECT Temp.Avg_DSLO FROM (SELECT ROUND(AVG(DaySinceLastOrder)) AS Avg_DSLO FROM customer_churn) AS Temp)WHERE DaySinceLastOrder IS NULL; ---------- (updated table below)----------

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

CustomerID	Churn	Tenure	Pref	Warehou	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatu	Number	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmour
50082	0	0	P...	NULL	Male	2	2	Mobile	4	Married	2	1	16	0	2	2	123
50083	0	NULL	C...	24	Fe...	3	3	Mobile	2	Single	2	0	13	1	1	7	122
50084	0	13	C...	15	Male	3	4	Laptop ...	2	Divorced	10	0	14	1	1	2	149
50085	0	19	M...	6	Male	3	3	Fashion	5	Married	8	0	16	0	1	3	194
50086	1	0	M...	10	Male	2	5	Fashion	2	Divorced	2	0	11	7	9	8	204
50087	0	3	C...	6	Male	2	4	Fashion	3	Single	2	0	22	2	5	8	185
50088	0	0	C...	26	Fe...	2	3	Laptop ...	3	Single	2	0	14	1	1	7	200
50089	0	24	M...	14	Male	2	4	Fashion	4	Divorced	6	0	16	1	4	5	225
50090	1	0	P...	NULL	Fe...	2	3	Mobile	5	Single	6	1	11	1	1	4	127
50091	0	14	M...	12	Fe...	3	3	Fashion	2	Single	4	1	18	0	NULL	0	236
50092	1	0	C...	32	Fe...	2	4	Laptop ...	5	Single	6	0	15	0	1	1	167
50093	1	NULL	P...	6	Male	2	3	Mobile	3	Single	1	0	14	0	1	5	121

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2) Impute mode for the following columns: Tenure, CouponUsed, OrderCount.

Tenure

Ans)select customerId, Tenure from customer_churn where Tenure is null;

Result Grid | Filter Rows:

customerId	Tenure
50002	NULL
50003	NULL
50007	NULL
50008	NULL
50010	NULL
50016	NULL
50023	NULL
50028	NULL
50035	NULL
50037	NULL
50048	NULL
50055	NULL

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select * from customer_churn ;

CustomerID	Churn	Tenure	Pref	Warehou	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatu	Number	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
50001	1	4	M...	6	Fe...	3	3	Laptop ...	2	Single	9	1	11	1	1	5	160
50002	1	NULL	P...	8	Male	3	4	Mobile	3	Single	7	1	15	0	1	0	121
50003	1	NULL	P...	30	Male	2	4	Mobile	3	Single	6	1	14	0	1	3	120
50004	1	0	P...	15	Male	2	4	Laptop ...	5	Single	8	0	23	0	1	3	134
50005	1	0	P...	12	Male	3	3	Mobile	5	Single	3	0	11	1	1	3	130
50006	1	0	C...	22	Fe...	3	5	Mobile ...	5	Single	2	1	22	4	6	7	139
50007	1	NULL	P...	11	Male	2	3	Laptop ...	2	Divorced	4	0	14	0	1	0	121
50008	1	NULL	P...	6	Male	3	3	Mobile	2	Divorced	3	1	16	2	2	0	123
50009	1	13	P...	9	Male	3	4	Mobile	3	Divorced	2	1	14	0	1	2	127
50010	1	NULL	P...	31	Male	2	5	Mobile	3	Single	2	0	12	1	1	1	123
50011	1	4	M...	18	Fe...	2	3	Others	3	Divorced	2	0	16	9	15	8	295
50012	1	11	M...	6	Male	3	4	Fashion	3	Single	10	1	13	0	1	0	154

customer_churn 30 | Apply

UPDATE customer_churn SET Tenure = (SELECT mode_value from (SELECT Tenure AS mode_value FROM customer_churn GROUP BY Tenure ORDER BY COUNT(*) DESC LIMIT 1)temp)WHERE Tenure IS NULL;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

CustomerID	Churn	Tenure	Pref	Warehou	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatu	Number	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
50001	1	4	M...	6	Fe...	3	3	Laptop ...	2	Single	9	1	11	1	1	5	160
50002	1	1	P...	8	Male	3	4	Mobile	3	Single	7	1	15	0	1	0	121
50003	1	1	P...	30	Male	2	4	Mobile	3	Single	6	1	14	0	1	3	120
50004	1	0	P...	15	Male	2	4	Laptop ...	5	Single	8	0	23	0	1	3	134
50005	1	0	P...	12	Male	3	3	Mobile	5	Single	3	0	11	1	1	3	130
50006	1	0	C...	22	Fe...	3	5	Mobile ...	5	Single	2	1	22	4	6	7	139
50007	1	1	P...	11	Male	2	3	Laptop ...	2	Divorced	4	0	14	0	1	0	121
50008	1	1	P...	6	Male	3	3	Mobile	2	Divorced	3	1	16	2	2	0	123
50009	1	13	P...	9	Male	3	4	Mobile	3	Divorced	2	1	14	0	1	2	127
50010	1	1	P...	31	Male	2	5	Mobile	3	Single	2	0	12	1	1	1	123
50011	1	4	M...	18	Fe...	2	3	Others	3	Divorced	2	0	16	9	15	8	295

CouponUsed

select * from customer_churn ;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

CustomerID	Churn	Tenure	Pref	Warehou	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatus	Number(	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
50058	1	4	P...	26	Fe...	3	3	Mobile	4	Single	2	1	22	0	1	2	131
50059	0	23	P...	6	Male	2	4	Mobile ...	3	Divorced	3	0	15	0	1	2	137
50060	0	26	P...	19	Fe...	2	2	Laptop ...	5	Married	2	0	11	1	1	3	145
50061	0	6	P...	20	Fe...	3	3	Mobile	4	Divorced	2	0	13	0	1	6	128
50062	0	13	M...	10	Fe...	2	3	Others	3	Divorced	2	0	14	NULL	9	9	290
50063	0	28	M...	20	Male	3	2	Others	4	Single	9	0	11	NULL	2	15	287
50064	0	7	C...	8	Male	2	4	Laptop ...	3	Divorced	2	0	17	0	1	9	195
50065	0	24	M...	9	Fe...	3	3	Grocery	4	Single	4	0	18	0	NULL	0	237
50066	0	25	M...	6	Male	3	4	Others	3	Divorced	1	0	16	1	11	9	300
50067	0	1	P...	15	Fe...	2	3	Mobile	4	Single	2	0	16	1	1	3	123
50068	1	0	M...	29	Male	2	4	Laptop ...	3	Divorced	4	1	13	1	1	1	159

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select customerId, CouponUsed from customer_churn where CouponUsed is null;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customerId	CouponUsed
50062	NULL
50063	NULL
50070	NULL
50113	NULL
50139	NULL
50146	NULL
50155	NULL
50156	NULL
50166	NULL
50190	NULL
50205	NULL
50255	NULL

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UPDATE customer_churn SET CouponUsed = (SELECT mode_value from (SELECT CouponUsed AS mode_value FROM customer_churn GROUP BY CouponUsed ORDER BY COUNT(*) DESC LIMIT 1)temp)WHERE CouponUsed IS NULL; select customerId, CouponUsed from customer_churn where CouponUsed is null;

Result Grid | Filter Rows: | Expo

customerId	CouponUsed

select * from customer_churn ;

CustomerID	Churn	Tenure	Pref	Warehou	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatus	Number(	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
50058	1	4	P...	26	Fe...	3	3	Mobile	4	Single	2	1	22	0	1	2	131
50059	0	23	P...	6	Male	2	4	Mobile ...	3	Divorced	3	0	15	0	1	2	137
50060	0	26	P...	19	Fe...	2	2	Laptop ...	5	Married	2	0	11	1	1	3	145
50061	0	6	P...	20	Fe...	3	3	Mobile	4	Divorced	2	0	13	0	1	6	128
50062	0	13	M...	10	Fe...	2	3	Others	3	Divorced	2	0	14	1	9	9	290
50063	0	28	M...	20	Male	3	2	Others	4	Single	9	0	11	1	2	15	287
50064	0	7	C...	8	Male	2	4	Laptop ...	3	Divorced	2	0	17	0	1	9	195
50065	0	24	M...	9	Fe...	3	3	Grocery	4	Single	4	0	18	0	NULL	0	237
50066	0	25	M...	6	Male	3	4	Others	3	Divorced	1	0	16	1	11	9	300
50067	0	1	P...	15	Fe...	2	3	Mobile	4	Single	2	0	16	1	1	3	123
50068	1	0	M...	29	Male	2	4	Laptop ...	3	Divorced	4	1	13	1	1	1	159

OrderCount.

select * from customer_churn ;

CustomerID	Churn	Tenure	Pref	Warehou	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatus	Number(	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
50058	1	4	P...	26	Fe...	3	3	Mobile	4	Single	2	1	22	0	1	2	131
50059	0	23	P...	6	Male	2	4	Mobile ...	3	Divorced	3	0	15	0	1	2	137
50060	0	26	P...	19	Fe...	2	2	Laptop ...	5	Married	2	0	11	1	1	3	145
50061	0	6	P...	20	Fe...	3	3	Mobile	4	Divorced	2	0	13	0	1	6	128
50062	0	13	M...	10	Fe...	2	3	Others	3	Divorced	2	0	14	1	9	9	290
50063	0	28	M...	20	Male	3	2	Others	4	Single	9	0	11	1	2	15	287
50064	0	7	C...	8	Male	2	4	Laptop ...	3	Divorced	2	0	17	0	1	9	195
50065	0	24	M...	9	Fe...	3	3	Grocery	4	Single	4	0	18	0	NULL	0	237
50066	0	25	M...	6	Male	3	4	Others	3	Divorced	1	0	16	1	11	9	300
50067	0	1	P...	15	Fe...	2	3	Mobile	4	Single	2	0	16	1	1	3	123
50068	1	0	M...	29	Male	2	4	Laptop ...	3	Divorced	4	1	13	1	1	1	159

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select customerId, OrderCount from customer_churn where OrderCount is null;

Result Grid | Filter Rows:

customerId	OrderCount
50065	NULL
50091	NULL
50111	NULL
50121	NULL
50144	NULL
50151	NULL
50171	NULL
50176	NULL
50192	NULL
50224	NULL
50266	NULL
50277	NULL

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UPDATE customer_churn SET OrderCount = (SELECT mode_value from (SELECT OrderCount AS mode_value FROM customer_churn GROUP BY OrderCount ORDER BY COUNT(*) DESC LIMIT 1)temp WHERE OrderCount IS NULL;

select * from customer_churn ;

CustomerID	Chur	Tenure	Pref	Wareho	Gend	HourSpe	Numb	PreferedO	Satis	MaritalStatu	Number	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
50058	1	4	P...	26	Fe...	3	3	Mobile	4	Single	2	1	22	0	1	2	131
50059	0	23	P...	6	Male	2	4	Mobile ...	3	Divorced	3	0	15	0	1	2	137
50060	0	26	P...	19	Fe...	2	2	Laptop ...	5	Married	2	0	11	1	1	3	145
50061	0	6	P...	20	Fe...	3	3	Mobile	4	Divorced	2	0	13	0	1	6	128
50062	0	13	M...	10	Fe...	2	3	Others	3	Divorced	2	0	14	1	9	9	290
50063	0	28	M...	20	Male	3	2	Others	4	Single	9	0	11	1	2	15	287
50064	0	7	C...	8	Male	2	4	Laptop ...	3	Divorced	2	0	17	0	1	9	195
50065	0	24	M...	9	Fe...	3	3	Grocery	4	Single	4	0	18	0	2	0	237
50066	0	25	M...	6	Male	3	4	Others	3	Divorced	1	0	16	1	11	9	300
50067	0	1	P...	15	Fe...	2	3	Mobile	4	Single	2	0	16	1	1	3	123
50068	1	0	M...	29	Male	2	4	Laptop ...	3	Divorced	4	1	13	1	1	1	159
50069	0	1	C...	13	Male	2	2	Mobile	2	Married	5	0	14	0	2	2	171

customer_churn 48

select customerId, OrderCount from customer_churn where OrderCount is null;

Result Grid | Filter Rows:

customerId	OrderCount

3)Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.

ANS) DELETE FROM customer_churn WHERE WarehouseToHome > 100

SELECT customerId, WarehouseToHome FROM customer_churn WHERE WarehouseToHome>100;

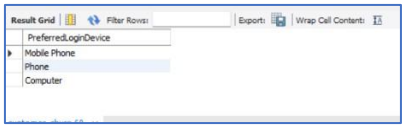
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customerId	WarehouseToHome

Dealing with Inconsistencies:

1) Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferedOrderCat' column with “Mobile Phone” to ensure uniformity.

Ans) SELECT DISTINCT PreferredLoginDevice FROM customer_churn;

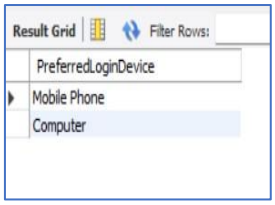


PreferredLoginDevice
Mobile Phone
Phone
Computer

UPDATE customer_churn SET PreferredLoginDevice = 'Mobile Phone' WHERE PreferredLoginDevice = 'Phone';

UPDATE customer_churn SET PreferredLoginDevice = 'Mobile Phone' WHERE PreferredLoginDevice = 'Mobile';

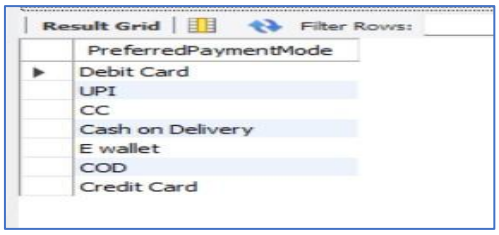
SELECT DISTINCT PreferredLoginDevice FROM customer_churn;



PreferredLoginDevice
Mobile Phone
Computer

2)Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.

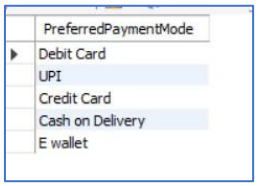
Ans) SELECT DISTINCT PreferredPaymentMode FROM customer_churn;



PreferredPaymentMode
Debit Card
UPI
CC
Cash on Delivery
E wallet
COD
Credit Card

UPDATE customer_churn SET PreferredPaymentMode = 'Cash on Delivery' WHERE PreferredPaymentMode = 'COD';

UPDATE customer_churn SET PreferredPaymentMode = 'Credit Card' WHERE PreferredPaymentMode = 'CC';



PreferredPaymentMode
Debit Card
UPI
Credit Card
Cash on Delivery
E wallet

Data Transformation

Column Renaming:

1) Rename the column "PreferedOrderCat" to "PreferredOrderCat".

2) Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

Ans) 1- ALTER TABLE customer_churn RENAME COLUMN PreferedOrderCat to PreferredOrderCat;

2- ALTER TABLE customer_churn RENAME COLUMN HourSpendOnApp TO HoursSpendOnAPP;

CustomerID	Churn	Tenure	PreferredLoginDevice	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpendOnAPP	Numb	PreferredOrderCat	Satis	MaritalStatu	Number(	Complain
50001	1	4	Mobile Phone	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single	9	1
50002	1	1	Mobile Phone	8	UPI	Male	3	4	Mobile	3	Single	7	1
50003	1	1	Mobile Phone	30	Debit Card	Male	2	4	Mobile	3	Single	6	1
50004	1	0	Mobile Phone	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single	8	0
50005	1	0	Mobile Phone	12	Credit Card	Male	3	3	Mobile	5	Single	3	0

Creating New Columns:

1)Create a new column named ‘ComplaintReceived’ with values "Yes" if the corresponding value in the ‘Complain’ is 1, and "No" otherwise.

ALTER TABLE customer_churn ADD ComplaintReceived VARCHAR(10);

UPDATE customer_churn SET ComplaintReceived = CASE WHEN Complain = 1 THEN 'YES' ELSE 'NO' END;

PreferredPaymentMode	Gender	HoursSpendOnAPP	Numb	PreferredOrderCat	Satis	MaritalStatu	Number(	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived
Debit Card	Female	3	3	Laptop & Accessory	2	Single	9	1	11	1	1	5	160	YES
UPI	Male	3	4	Mobile	3	Single	7	1	15	0	1	0	121	YES
Debit Card	Male	2	4	Mobile	3	Single	6	1	14	0	1	3	120	YES
Debit Card	Male	2	4	Laptop & Accessory	5	Single	8	0	23	0	1	3	134	NO
Credit Card	Male	3	3	Mobile	5	Single	3	0	11	1	1	3	130	NO

2) Create a new column named 'ChurnStatus'. Set its value to “Churned” if the corresponding value in the 'Churn' column is 1, else assign “Active”.

Ans) ALTER TABLE customer_churn ADD ChurnStatus VARCHAR(10);

UPDATE customer_churn SET ChurnStatus = CASE WHEN Churn = 1 THEN 'Churned' ELSE 'Active' END;

edPaymentMode	Gender	HoursSpendOnAPP	Numb	PreferredOrderCat	Satis	MaritalStatu	Number(	Complain	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived	ChurnStatus
rd	Female	3	3	Laptop & Accessory	2	Single	9	1	11	1	1	5	160	YES	Churned
	Male	3	4	Mobile	3	Single	7	1	15	0	1	0	121	YES	Churned
rd	Male	2	4	Mobile	3	Single	6	1	14	0	1	3	120	YES	Churned
rd	Male	2	4	Laptop & Accessory	5	Single	8	0	23	0	1	3	134	NO	Churned
ard	Male	3	3	Mobile	5	Single	3	0	11	1	1	3	130	NO	Churned

Column Dropping:

1) Drop the columns "Churn" and "Complain" from the table.

ALTER TABLE customer_churn DROP COLUMN Complain;

ALTER TABLE customer_churn DROP COLUMN Churn;

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpendOnAPP	Numb	PreferredOrderCat	Satisfactio	MaritalStatu	Number(	OrderAmount	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ComplaintReceived	ChurnSta
50001	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2	Single	9	11	1	1	5	160	YES	Churned
50002	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile	3	Single	7	15	0	1	0	121	YES	Churned
50003	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile	3	Single	6	14	0	1	3	120	YES	Churned
50004	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5	Single	8	23	0	1	3	134	NO	Churned
50005	0	Mobile Phone	1	12	Credit Card	Male	3	3	Mobile	5	Single	3	11	1	1	3	130	NO	Churned

Data Exploration and Analysis:

1)Retrieve the count of churned and active customers from the dataset.

Ans) SELECT ChurnStatus, COUNT(*) FROM customer_churn GROUP BY ChurnStatus;

Result Grid | Filter Rows:

ChurnStatus	COUNT(*)
Churned	948
Active	4680

2)Display the average tenure and total cashback amount of customers who churned.

Ans) SELECT AVG(Tenure) AS AvgTenure,SUM(CashbackAmount) AS TotalCashback FROM customer_churn WHERE ChurnStatus = 'Churned';

Result Grid | Filter Rows:

AvgTenure	TotalCashback
3.1762	152030

3)Determine the percentage of churned customers who complained.

Ans) SELECT(sum(CASE WHEN ComplaintReceived = 'YES' THEN 1 END) * 100 / COUNT(*))

AS Percentage_Complained FROM customer_churn WHERE ChurnStatus = 'Churned';

Result Grid | Filter Rows:

Percentage_Complained
53.5865

4) Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

Ans) SELECT CityTier,COUNT(*) AS churned_customercount FROM customer_churn

WHERE ChurnStatus = 'Churned' AND PreferredOrderCat = 'Laptop & Accessory'

GROUP BY CityTier ORDER BY churned_customercount DESC LIMIT 1;

Result Grid | Filter Rows:

CityTier	churned_customercount
3	150

5)Identify the most preferred payment mode among active customers.

Ans) SELECT PreferredPaymentMode, COUNT(*) as count_mode FROM customer_churn WHERE ChurnStatus='Active' GROUP BY PreferredPaymentMode ORDER BY COUNT(*) DESC LIMIT 1;

Result Grid | Filter Rows:

PreferredPaymentMode	COUNT(*)
Debit Card	1956

6)Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

Ans) SELECT SUM(OrderAmountHikeFromlastYear) AS Total_orderhike_from_lastyear

FROM customer_churn WHERE MaritalStatus = 'Single' AND PreferredLoginDevice = 'Mobile Phone';

Result Grid | Filter Rows:

Total_orderhike_from_lastyear
19593

7)Find the average number of devices registered among customers who used UPI as their

preferred payment mode. Ans) SELECT AVG(NumberOfDeviceRegistered) AS Avg_devices

FROM customer_churn WHERE

PreferredPaymentMode = 'UPI';

Result Grid

Avg_devices
3.7174

9) Determine the city tier with the highest number of customers.

Ans) SELECT CityTier,COUNT(*) AS customer_count FROM customer_churn GROUP BY CityTier ORDER BY customer_count DESC LIMIT 1;

Result Grid | Filter Rows:

CityTier	customer_count
1	3666

10) Identify the gender that utilized the highest number of coupons.

Ans) SELECT Gender, SUM(CouponUsed) AS Total_coupon_used FROM customer_churn GROUP BY Gender ORDER BY Total_coupon_used DESC LIMIT 1;

Result Grid | Filter Rows:

Gender	Total_coupon_used
Male	5629

11)List the number of customers and the maximum hours spent on the app in each preferred order category.

Ans) SELECT PreferredOrderCat,COUNT(DISTINCT CustomerID) AS Number_of_customers,MAX(HoursSpendOnApp)
AS Max_hours_spent FROM customer_churn GROUP BY PreferredOrderCat ORDER BY PreferredOrderCat;

PreferredOrderCat	Number_of_customers	Max_hours_spent
Fashion	826	5
Grocery	410	4
Laptop & Accessory	2050	5
Mobile	808	4
Mobile Phone	1270	5
Others	264	4

➢12)Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.

Ans) SELECT SUM(OrderCount) AS total_order_count FROM customer_churn WHERE PreferredPaymentMode = 'Credit Card' AND SatisfactionScore = (SELECT MAX(SatisfactionScore) FROM customer_churn WHERE PreferredPaymentMode = 'Credit Card');

Result Grid

total_order_count
1122

➢13)What is the average satisfaction score of customers who have complained?

Ans) SELECT AVG(SatisfactionScore) AS average_satisfaction_score FROM customer_churn WHERE ComplaintReceived ='YES';

Result Grid Filter Rows:

average_satisfaction_score
2.9988

14) List the preferred order category among customers who used more than 5coupons.

Ans) SELECT PreferredOrderCat AS Order_category,COUNT(*) AS customer_count FROM customer_churn WHERE CouponUsed >5 GROUP BY PreferredOrderCat ORDER BY customer_count DESC;

Order_category	customer_count
Laptop & Accessory	99
Fashion	89
Grocery	42
Mobile Phone	30
Others	28
Mobile	15

15)List the top 3 preferred order categories with the highest average cashback amount.

Ans) SELECT PreferredOrderCat,AVG(CashbackAmount) AS average_cashback FROM customer_churn GROUP BY PreferredOrderCat ORDER BY average_cashback DESC LIMIT 3;

Result Grid Filter Rows:

PreferredOrderCat	average_cashback
Others	304.4545
Grocery	266.2366
Fashion	210.4031

16) Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

Ans) SELECT PreferredPaymentMode FROM customer_churn GROUP BY PreferredPaymentMode HAVING round(AVG(Tenure)) = 10 AND SUM(OrderCount) > 500;

PreferredPaymentMode
Debit Card
Credit Card
E wallet

17) Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

Ans)
```
SELECT
  CASE
    WHEN WarehouseToHome <= 5 THEN 'Very Close Distance'
    WHEN WarehouseToHome <= 10 THEN 'Close Distance'
    WHEN WarehouseToHome <= 15 THEN 'Moderate Distance'
    ELSE 'Far Distance'
END AS distance,
  ChurnStatus,
  COUNT(*) AS customers
FROM customer_churn
GROUP BY distance, ChurnStatus
ORDER BY
  MIN(WarehouseToHome),
  ChurnStatus DESC;
```

Result Grid | Filter Rows: | Exp

distance	ChurnStatus	customers
Very Close Distance	Churned	1
Very Close Distance	Active	7
Close Distance	Churned	265
Close Distance	Active	1696
Moderate Distance	Churned	184
Moderate Distance	Active	1106
Far Distance	Churned	498
Far Distance	Active	1871

18) List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

Ans)
```
SELECT CustomerID,OrderCount,PreferredOrderCat,PreferredPaymentMode,CashbackAmount
FROM customer_churn WHERE MaritalStatus = 'Married'AND  CityTier = 1 AND
 OrderCount > (SELECT AVG(OrderCount)  FROM customer_churn) ORDER BY OrderCount DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap C

CustomerID	OrderCount	PreferredOrderCat	PreferredPaymentMode	CashbackAmount
53605	16	Grocery	Credit Card	261
53628	16	Grocery	Credit Card	296
54206	16	Others	Cash on Delivery	320
54287	16	Fashion	Credit Card	228
54586	16	Grocery	Debit Card	295
55075	16	Grocery	Credit Card	261
55098	16	Grocery	Credit Card	296
50813	15	Grocery	Credit Card	271
51391	15	Others	Cash on Delivery	295
51472	15	Fashion	Credit Card	203

19) List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

a) Create a 'customer_returns' table in the 'ecomm' database and insert the following data:

Ans)

```
CREATE TABLE customer_returns (
   ReturnID INT PRIMARY KEY,
   CustomerID INT NOT NULL,
   ReturnDate DATE NOT NULL,
   RefundAmount DECIMAL(10, 2) NOT NULL
);

INSERT INTO customer_returns (ReturnID, CustomerID, ReturnDate, RefundAmount)
VALUES
   (1001, 50022, '2023-01-01', 2130.00),
   (1002, 50316, '2023-01-23', 2000.00),
   (1003, 51099, '2023-02-14', 2290.00),
   (1004, 52321, '2023-03-08', 2510.00),
   (1005, 52928, '2023-03-20', 3000.00),
   (1006, 53749, '2023-04-17', 1740.00),
```

```
(1007, 54206, '2023-04-21', 3250.00),
(1008, 54838, '2023-04-30', 1990.00); SELECT
* FROM customer_returns;
```

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130.00
1002	50316	2023-01-23	2000.00
1003	51099	2023-02-14	2290.00
1004	52321	2023-03-08	2510.00
1005	52928	2023-03-20	3000.00
1006	53749	2023-04-17	1740.00
1007	54206	2023-04-21	3250.00
1008	54838	2023-04-30	1990.00

b) Display the return details along with the customer details of those who have churned and have made complaints.

```
SELECT Ret.*,
customer.CityTier,customer.PreferredOrderCat,customer.PreferredPaymentMode
FROM customer_returns Ret JOIN customer_churn customer ON ret.CustomerID = customer.CustomerID
WHERE customer.ChurnStatus = 'Churned' AND customer.ComplaintReceived = 'Yes';
```

ReturnID	CustomerID	ReturnDate	RefundAmount	CityTier	PreferredOrderCat	PreferredPaymentMode
1002	50316	2023-01-23	2000.00	2	Fashion	UPI
1004	52321	2023-03-08	2510.00	3	Grocery	E wallet
1006	53749	2023-04-17	1740.00	3	Laptop & Accessory	Credit Card